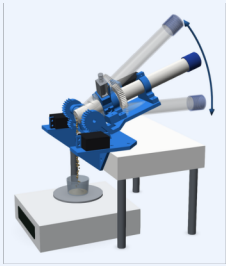
Cheap: measured on the doser itself

Physics-based ML

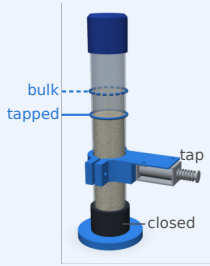
surrogate models and DEM simulations find the links



Feed factor

mg per auger revolution at several tilts, and before/after humid air

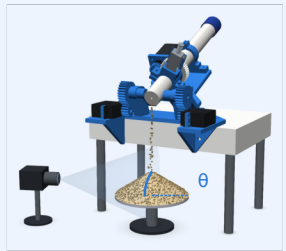
measured now (13 powders)



“Ghost dose”

transparent doser, outlet closed, tapped: bulk and tapped density

proposed: thin-walled SLA vessel



Angle of repose

camera on the pile the doser pours

proposed: camera at the outlet

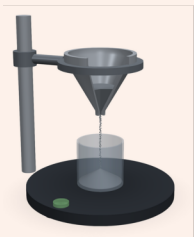


State and history of the powder

composition; how it was made, sieved, dried, stored or reused

infer the properties of a new powder

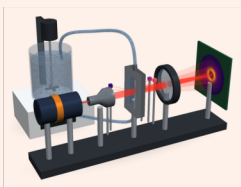
known properties speed up calibration



Hall/Carney flow

50 g through a standard funnel (ASTM B213, B964)

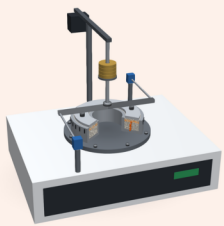
in-house: funnel set



Particle size (PSD)

SEM images (BSE mode) or laser diffraction

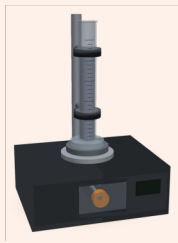
in-house SEM, or outside lab



Shear cell, ff_c

cohesion under load (ASTM D6773, D7891)

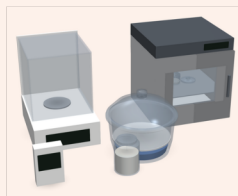
outside lab



Tapped density

graduated cylinder, loose then tapped (ASTM B527)

in-house: cylinder



Moisture

loss on drying (weigh, dry, reweigh)

in-house: oven, balance

Standard tests: a separate instrument for each, some only in outside labs